

Article

Research on Trajectory Prediction Algorithm Based on Unmanned Aerial Vehicles Behavioral Intentions

Yi Cao ¹, Jiandong Zhang ¹, Guoqing Shi ^{1,*}, Qiming Yang ^{1,*} and Chengbiao Zhang ²¹ School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710129, China; caoyi@mail.nwpu.edu.cn (Y.C.); jdzhang@nwpu.edu.cn (J.Z.)² Jiangxi Hongdu Aviation Industry Co., Ltd., Nanchang 330024, China; zhangcb007@avic.com

* Correspondence: shiguqing@nwpu.edu.cn (G.S.); yangqm@nwpu.edu.cn (Q.Y.)

Abstract

In the unmanned aerial vehicles (UAVs) flight control and navigation guidance system, trajectory prediction serves as a critical foundational component, with its accuracy and reliability directly influencing the system performance of the UAVs. However, existing research has predominantly focused on optimizing algorithm efficiency, failing to fully consider the impact of the UAV's flight status on its trajectory. This has resulted in significant discrepancies between predicted results and actual trajectories in complex scenarios. Therefore, this paper proposes a trajectory prediction algorithm that integrates the UAVs' behavioral intentions. Firstly, a behavioral intention recognition model is constructed using the Support Vector Machine (SVM) to accurately discriminate the UAV's motion patterns and output the probability distribution of its future actions, thereby integrating semantic-level intention information into the prediction process. Secondly, the Bidirectional Gated Recurrent Unit (Bi-GRU) is employed to mine the spatial-temporal correlation features from trajectory data. Additionally, an attention mechanism is introduced to capture key information of sequence, enhancing the model's ability to represent complex motion trends. The results of simulation experiments demonstrate that this algorithm exhibits significant advantages in terms of trajectory prediction accuracy and scene adaptability, providing more practical technical support for intelligent navigation and safety control of UAVs.



Academic Editor: Samer Hanoun

Received: 23 July 2025

Revised: 7 September 2025

Accepted: 11 September 2025

Published: 12 September 2025

Citation: Cao, Y.; Zhang, J.; Shi, G.; Yang, Q.; Zhang, C. Research on Trajectory Prediction Algorithm Based on Unmanned Aerial Vehicles Behavioral Intentions. *Drones* **2025**, *9*, 640. <https://doi.org/10.3390/drones9090640>

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Keywords: UAV; support vector machine; random forest; Bi-GRU; attention; neural networks

1. Introduction

With the continuous advancement of aviation technology, communication technology, electronic technology and computer science, UAV technology [1] has achieved breakthrough progress. Leveraging its exceptional flexibility and efficiency, UAVs have transitioned from specialized applications to diverse scenarios, finding widespread use in military strikes [2–5], disaster relief [6,7], agricultural irrigation [8,9], environmental monitoring [10,11], logistics delivery [12,13] and various other fields, emerging as an innovative force that is transforming production and lifestyles.

However, with the soaring number of UAVs and the expansion of their application scenarios, their safety issues have become increasingly prominent [14]. To ensure UAVs operate in compliance with regulations in complex environments and avoid potential risks, predict precise-trajectory technology [15–17] has emerged as a core capability. By predicting the flight path and heading direction of UAVs in advance, we can achieve real-time risk

warning, dynamic airspace management and control, as well as emergency intervention, thereby establishing a technological safeguard for the safe operation of UAVs.

Currently, most UAV trajectory prediction algorithms can be broadly classified into two categories: filter estimation [18] and neural networks [19,20]. The modern development and widespread application of filter estimation originated with the Kalman filter algorithm (KF) [21]. Its first engineering application can be traced back to the Apollo Moon landing program in 1961. This algorithm is based on the minimum mean square error criterion and achieves recursive state estimation in Gaussian noise environments. Zhu et al. [22] designed a Recurrent Neural Network-enhanced Kalman Filter (RNN-KF) algorithm for UAV trajectory tracking using Automatic Dependent Surveillance-Broadcast (ADS-B) data. This algorithm leverages the RNN to capture UAV maneuvering behaviors and data noise levels, thereby assisting in adaptively adjusting the process noise matrix and observation noise matrix of the KF. This approach addresses the impact of transmission noise on tracking accuracy, providing an effective solution for precise UAV tracking and airspace safety assurance when the UAV is beyond communication range. Hematulin et al. [23] proposed an integrated KF-based solution to address the collision avoidance problem in autonomous trajectory planning for multiple UAV. This method constructs a measurement model using GPS and magnetometer data, achieving UAV state estimation through the integrated KF. Experimental results demonstrate its outstanding performance even in complete mission scenarios involving fused positioning and rendezvous operations. Ma et al. [24] enhanced the success rate of capturing illegal drones through human-involved prediction of their flight trajectories, the Extended Kalman Filter algorithm (EKF) is integrated into an Image-based Visual Servoing (IBVS) system for the purpose of estimating the target drone's future positions. Yang et al. [25] addressed the positioning challenge faced by drones in extremely confined environments due to satellite signal blockage by employing an adaptive EKF to fuse data from an Inertial Measurement Unit (IMU) and Ultraband (UWB) systems. By adaptively adjusting the noise covariance matrix using historical estimates and recorded information, the adverse effects of measurement-noise oscillations on positioning stability in time-varying environments are effectively mitigated.

In recent years, neural networks have gradually become the mainstream solution for trajectory prediction problems due to their powerful capability in fitting nonlinear functions. Zhang et al. [26] utilized the SVM model to distinguish different flight states of drones, they constructed corresponding neural network models, respectively, for trajectory prediction and validated the effectiveness of the algorithm through real flight data. Zhong et al. [27] proposed a short-term four-dimensional (4D) trajectory prediction method based on spatiotemporal trajectory clustering. This approach extracts trajectory label information through spatiotemporal trajectory clustering and Convolutional Neural Network (CNN), and employs Long Short-Term Memory (LSTM) networks to achieve high-precision prediction of drones with stochastic behavioral intentions. Song et al. [28] proposed a trajectory planning and tracking method for UAV swarms based on Deep Neural Network (DNN)-enhanced Model Predictive Control (MPC). By integrating DNN with the MPC algorithm, this approach addresses large-scale, multi-constrained optimization problems in UAV swarm control, overcoming the limitation of traditional MPC that relies on accurate prior models. Xie et al. [29] proposed a framework for UAV trajectory prediction based on Gaussian Process Regression (GPR). This framework effectively addresses the impact of external environmental factors on UAV trajectory prediction and enables online real-time trajectory forecasting. The effectiveness of the proposed framework has been verified through both simulation and real-world datasets. Tan et al. [30] investigated eight typical maneuver types and conducted offline training of an LSTM network to achieve accurate classification of trajectory types and develop an online prediction approach.

According to the aforementioned literature, existing research in the field of UAV trajectory prediction has already achieved certain accomplishments and accumulated a wealth of knowledge, yet it exhibits a distinct technological bias overall. The majority of studies concentrate on the temporal evolution of trajectory sequences, aiming to establish a direct mapping from historical UAV trajectory data to future trajectories through predictive algorithms. Significant efforts have been devoted to the iterative optimization of these algorithms in such research.

However, existing research has notable limitations: it generally treats UAVs as “black-box trajectory carriers,” overlooking the impact of the UAVs’ flight states on their trajectories. In reality, the current flight state of a UAV influences, to a certain extent, the evolving trend of its trajectory. This makes it difficult for traditional models to accurately capture trajectory characteristics in complex environments, resulting in discrepancies between predicted outcomes and actual flight scenarios.

To address this issue, this paper incorporates the maneuver intention of UAVs into the trajectory prediction model. The main innovations are as follows:

- (1) Design a collaborative framework integrating intention recognition and trajectory prediction. This framework extracts the flight status of UAVs in real time to predict their current maneuver intentions, and then incorporates these intentions as key semantic constraints into the trajectory prediction algorithm to achieve accurate trajectory forecasting.
- (2) Construct a multi-class SVM model for maneuver intention recognition. To address the issue of redundant interference in high-dimensional flight state information, the random forest algorithm is employed for feature importance evaluation and selection. A probabilistic output mechanism is introduced to transform the discrete SVM class determinations into probability distributions of various maneuver modes, thereby providing prior information for the subsequent trajectory prediction model.
- (3) Design an attention-based Bi-GRU neural network prediction algorithm. The Bi-GRU network is utilized to capture both forward and backward dependencies within sequences, enabling the construction of bidirectional representations of spatiotemporal trajectory features. An attention mechanism is introduced to focus on key features within the trajectory data, significantly enhancing adaptability and prediction accuracy in complex task scenarios.

The subsequent sections of this paper are structured as follows: Section 2 establishes a six-degree-of-freedom (6-DOF) dynamics model for UAV and overall framework description. Section 3 introduces the random forest algorithm and SVM model, achieving intention recognition for UAV maneuvers. Section 4 presents the Bi-GRU and attention mechanism and designs a neural network-based trajectory prediction algorithm. Section 5 employs comparative experiments to validate the reliability of the proposed algorithm. Section 6 summarizes the entire paper and explores future research directions.

2. Flight Dynamic Model and Overall Framework Description

2.1. Flight Dynamic Model

To achieve accurate trajectory prediction for fixed-wing UAV, this paper constructs a six-degree-of-freedom (6-DOF) dynamics model [31] that more closely aligns with actual flight scenarios, based on the motion characteristics and flight laws of fixed-wing UAVs. This model precisely simulates core state information such as attitude, velocity, and position of the UAV under different flight conditions, providing a basis for subsequent trajectory prediction. The model encompasses three translational degrees of freedom (displacement

along the X, Y, and Z axes) and three rotational degrees of freedom (roll, pitch, and yaw). The acceleration formulas, fixed within the body-fixed coordinate system, are as follows:

$$\begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} = \begin{bmatrix} u \\ v \\ w \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} + \frac{1}{m} \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} + C_g^b \begin{bmatrix} 0 \\ 0 \\ G \end{bmatrix} \quad (1)$$

Here, u, v, w respectively represent the velocities along the x, y, z axes of the body-fixed coordinate system. p, q, r denote the angular velocities about the three axes. F_x, F_y, F_z represent the external forces acting along the three axes, m denotes the mass of the aircraft, G represents the acceleration due to gravity, and C signifies the transformation matrix from the ground coordinate system to the body-fixed coordinate system.

$$\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = I^{-1} \left(- \begin{bmatrix} p \\ q \\ r \end{bmatrix} + I \begin{bmatrix} p \\ q \\ r \end{bmatrix} + \begin{bmatrix} M_x \\ M_y \\ M_z \end{bmatrix} \right) \quad (2)$$

The angular accelerations of the body-fixed coordinate system are given by Formula (2), where I represents the moment of inertia of the aircraft, and M_x, M_y, M_z respectively denote the external moments along the body-fixed coordinate system.

The attitude changes and position movements of the UAV in space can all be characterized by the parameterization generated in the body coordinate system and the ground coordinate system through a six-degree-of-freedom model. It is the temporal correlation of these quantitative parameters that generates the trajectory data of the UAV. In this section, based on the temporal characteristics of the trajectory data, a basic flight maneuver action library S for the UAV is constructed. Each type of maneuver action V_C is clearly defined as a set of continuous state parameters, and its mathematical expression can be defined as:

$$S = \{V_{C1}, V_{C2}, \dots, V_{Cn}\} \quad (3)$$

$$V_{Ci} = \{x_{Ci}^{(t_1)}, x_{Ci}^{(t_2)}, \dots, x_{Ci}^{(t_n)}\} \quad (4)$$

Among them, $x_C^{(t_i)}$ is the instantaneous flight information at time t_i and V_{Ci} is the i -th type of maneuver action.

Specifically, this action library includes four types of basic actions: level flight, turn, climb, and dive, as shown in Figure 1. By analyzing the change trends of flight parameters at continuous moments, accurate definitions of different maneuver actions are achieved. These classifications cover the typical maneuver modes in various tasks of fixed-wing UAVs and provide a clear semantic label system for the maneuver intention recognition model.

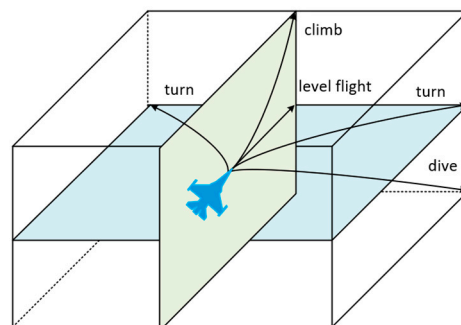


Figure 1. UAV maneuver action library.

2.2. Overall Framework Description

Traditional trajectory prediction treats the UAV as a “black box” and extrapolates future positions solely from historical flight paths. Such approaches fail to capture the influence of maneuver behaviors on trajectory evolution, leading to large errors in complex airspace. To address this limitation, this paper proposes a cascaded scheme that first recognizes maneuver intention and then performs trajectory prediction, dividing the entire task into two successive modules. As illustrated in Figure 2, the framework consists of two cascaded stages:

- (1) **Maneuver-intention recognition stage**—responsible for extracting maneuver intention. This stage takes real-time flight states as input, performs feature selection and probabilistic SVM discrimination, and outputs the posterior probability distribution over maneuver actions, thereby providing a quantitative representation of the UAV’s short-term behavioral intent. The detailed algorithmic flow is presented in Section 3.
- (2) **Trajectory-prediction stage**—responsible for spatio-temporal modeling. While encoding historical trajectories, this stage embeds the intention probabilities produced by the upper stage and utilizes a deep sequential network to achieve end-to-end future trajectory prediction. The network architecture and training strategy are elaborated in Section 4.

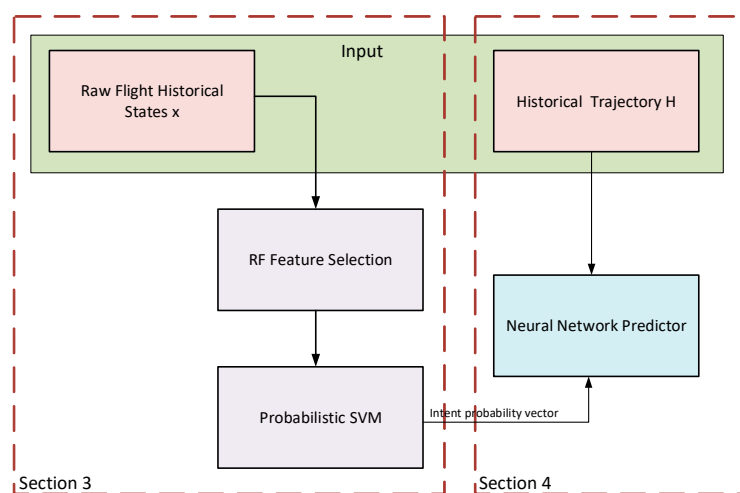


Figure 2. Framework description.

Information between the two stages is transmitted only through the intention-probability vector in a unidirectional manner, without sharing any internal parameters or gradients. This loosely coupled design offers two key advantages: first, each module can be upgraded independently—if a superior trajectory network emerges later, it can replace the existing one without retraining the intention model; second, by concatenating the UAV maneuver intention probabilities with the trajectory, the network gains prior-aware inputs, which experimentally improve prediction accuracy.

3. Design of UAV Maneuver Intention Recognition Algorithm

The maneuvering intention of the UAV refers to the movement goals and behavior trends shown during its flight, and is the overall description of the current flight state. Based on the maneuver action library constructed in Section 2.1, this paper proposes an intention recognition algorithm, the process of which is connected by two major links: first, Section 3.1 uses random forest to measure and screen the importance of high-dimensional flight features, and retains the subset of features most relevant to maneuver intention; Then, in Section 3.2, a multi-class support vector machine is constructed, and Platt Scaling

is introduced to transform the discriminant result into a posterior probability distribution of maneuvering actions, providing interpretable semantic constraints for subsequent trajectory prediction.

3.1. Random Forest Algorithm

In the task of UAV maneuver intention recognition, the quality of features directly determines the performance of the classification model. The original state information of the UAV has a relatively high dimension, covering various features such as attitude angles and motion parameters. Among these features, there are not only key information highly related to the maneuver intention but also redundant components due to the correlation overlap between features. If the high-dimensional original data is directly input into the Support Vector Machine (SVM) for classification, redundant features may lead to a decline in the generalization ability of the model, thereby affecting the classification effect. For this reason, this section introduces the random forest algorithm to extract the key features of the UAV.

Random forest is an ensemble learning algorithm [32]. It trains multiple decision tree estimators in parallel, comprehensively outputs prediction results, and has the function of feature selection. As shown in Figure 3, for the UAV flight trajectory prediction task, let the training dataset be $D = \{x_1, x_2, \dots, x_n\}$, where x is a vector containing flight states and trajectory features. The Bootstrap Sampling method is used to randomly extract k times from the original dataset D to obtain sample subsets S_i of size M . Then, each sample subset S_i can be expressed as:

$$S_i = \{x_{i1}, x_{i2}, \dots, x_{iM}\} \quad (5)$$

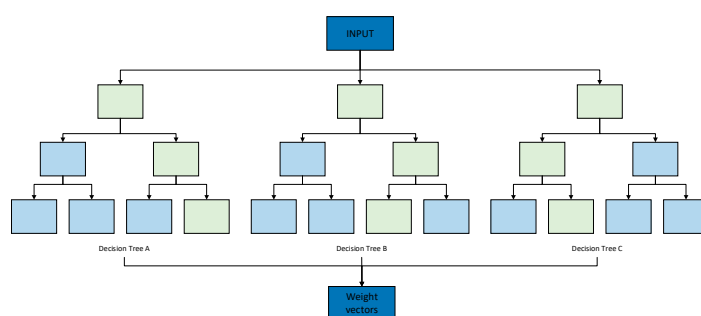


Figure 3. Random forest.

This sampling allows repeated samples, so that each subset not only retains the distribution of the original data but also introduces differences.

When training a single decision tree, to enhance the diversity of base classifiers and avoid overfitting, the random forest introduces random feature selection in the node splitting stage. Assume that the original feature set is $F = \{f_1, f_2, \dots, f_p\}$. Each time a node is split, m -dimensional features are randomly selected (usually $m = \log_2 p$) to form a candidate feature subset v , which can be expressed as:

$$v = \{f_{j_1}, f_{j_2}, \dots, f_{j_m}\} \subseteq F \quad (6)$$

f_{j_i} represents feature information, and F is the feature set. This strategy restricts the available features of a single node, forcing different decision trees to focus on the differences in feature combinations, thereby improving the adaptability of the integrated model to high-dimensional sparse data.

To quantify the discriminative value of features for classification results, Information Gain is used to calculate the contribution of each feature in the decision tree. Let D_{node} be

the sample subset at the current node, v the candidate feature subset, and candidate feature $f \in v$, the formula can be expressed as:

$$IG(D_{node}, f) = E(D_{node}) - \sum_{s \in Val(f)} \frac{|D_{node,s}|}{|D_{node}|} \times E(D_{node,s}) \quad (7)$$

Here, $E(\bullet)$ denotes the information entropy that measures the uncertainty of class labels. $Val(f)$ is the set of candidate split points derived after sorting the values of feature f , and $D_{node,v}$ is the sample subset of D_{node} that satisfies the splitting condition at split point s .

The greater the information gain, the more significantly the uncertainty of sample categories is reduced after splitting with this feature, and the stronger the discriminative ability of the feature. When splitting a decision tree node, all features in the candidate feature subset v are traversed, and the feature with the maximum information gain is selected for splitting.

In this research, a total of 18-dimensional UAV flight state features are collected, including three-dimensional position, three-dimensional velocity, three-dimensional acceleration, three-axis attitude angles (pitch, roll, yaw), three-axis angular rates, flight altitude, altitude change rate, and flight time. Due to the significant differences in the physical dimensions of each feature, all inputs are first standardized to eliminate the influence of dimensions. The formula can be expressed as:

$$X_{scaled}(i, j) = \frac{X(i, j) - \mu_j}{\sigma_j} \quad (8)$$

σ_j and μ_j are obtained from the statistics of the j -th feature across the training samples.

On 100 groups of samples, the importance of each feature is calculated according to the information gain criterion, and screening is performed with an 80% cumulative contribution rate as the threshold. As shown in Figure 4, 10 key features such as three-axis acceleration, pitch angle, three-axis velocity, roll angle, yaw angle change rate, and altitude change rate can explain 82.4% of the total information gain, significantly compressing the input dimension. At the same time, it provides representative core discriminative information for subsequent UAV flight intention recognition tasks.

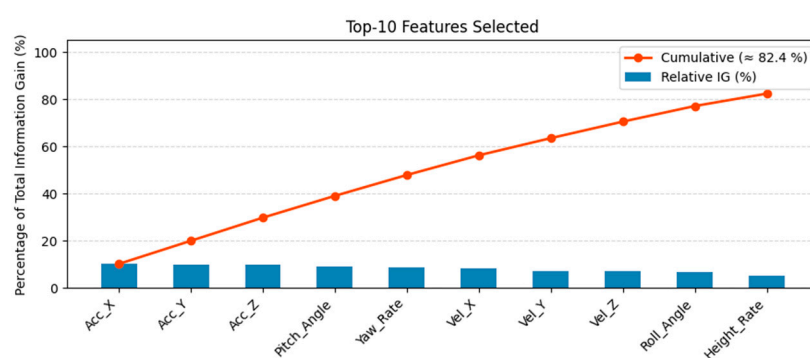


Figure 4. Features selected.

3.2. Support Vector Machine

SVM is a classic machine learning method proposed in 1995 by reference [33] based on statistical learning theory. Its core idea is to map the original training sample set $\{X, Y\}$ to a high-dimensional feature space through a kernel function, and then construct an optimal decision hyperplane to minimize the classification error.

Suppose the training set $\{X, Y\}$ is binary classification data, where the value of Y is $y_i \in \{+1, -1\}$. SVM constructs a linear regression function through nonlinear mapping. The formula can be expressed as:

$$\tilde{g}(x) = \omega \cdot x + b \quad (9)$$

Taking Figure 5 as an example, the two types of scatters (blue and brown) visually present the sample distribution, the solid line is the optimal decision hyperplane, and the distance γ between the dotted line and the hyperplane is the shortest distance from all datasets to the hyperplane, and the formula can be expressed as:

$$\gamma_i = \frac{\|\omega^T x_i + b\|}{\|\omega\|} \quad (i = 1, 2, \dots, N) \quad (10)$$

$$\gamma = \min\{\gamma_1, \gamma_2, \dots, \gamma_n\} \quad (11)$$

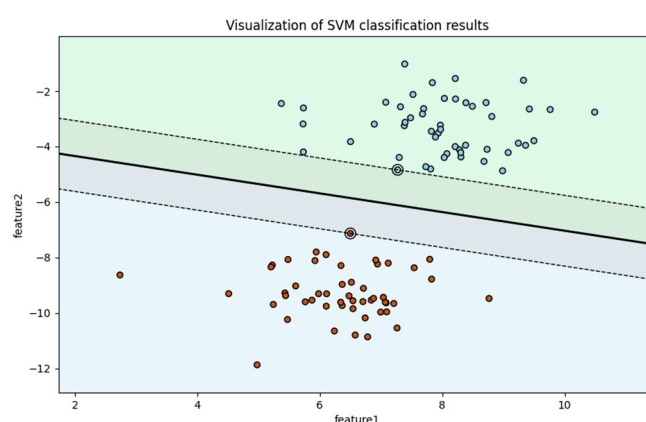


Figure 5. SVM algorithm.

For linearly separable datasets, the optimization goal of SVM is to maximize the classification margin. By incorporating the hyperplane parameter ω into the above formula, the basic optimization function of SVM can be derived as:

$$\begin{cases} \min \frac{1}{2} \|\omega\|^2 \\ s.t. y_i(\omega^T x_i + b) \geq 1 \end{cases} \quad (12)$$

To improve the generalization ability of the model, slack variables ξ_i and ξ_i^* and a regularization parameter C are usually introduced. Among them, ξ_i and ξ_i^* allow some samples to break through the constraint of the margin γ , thereby finding a balance between maximizing the margin and tolerating minor sample deviations, and reducing the overfitting risk of the model to noisy samples. The core function of the regularization term is to limit the norm of the weight vector ω . By penalizing excessively large weight parameters, it balances the relationship between margin maximization and classification errors. Therefore, the optimization problem can be transformed into solving the minimum value of the following formula:

$$\begin{cases} \min \frac{1}{2} \|\omega\|^2 + C \sum_{i=1}^N (\xi_i + \xi_i^*), \\ y_i - \omega x_i - b \leq \gamma + \xi_i \\ s.t. \omega x_i + b - y_i \leq \gamma + \xi_i^* \\ \xi_i \geq 0, \xi_i^* \geq 0, i = 1, 2, \dots, n \end{cases} \quad (13)$$

To effectively handle the nonlinear relationships in the data, this paper adopts the Gaussian kernel function to achieve feature space mapping. Its expression is as follows:

$$k(x_i, x) = \exp(-\|x_i - x\|^2 / \sigma^2) \quad (14)$$

Here, σ is a parameter of the Gaussian kernel function.

Although the UAV flight maneuver library has been constructed, the actual flight trajectory is usually composed of the combination of multiple basic maneuver actions. As shown in Figure 6, UAV can simultaneously perform a turn during the climb process, so trajectory analysis needs to judge multi-category composite maneuvers such as “turn + climb”. Since the traditional SVM is only suitable for binary classification tasks, this section introduces the One-vs-All (OvA) strategy to expand the multi-classification capability of SVM, so as to meet the classification requirements of complex maneuver actions of UAV.

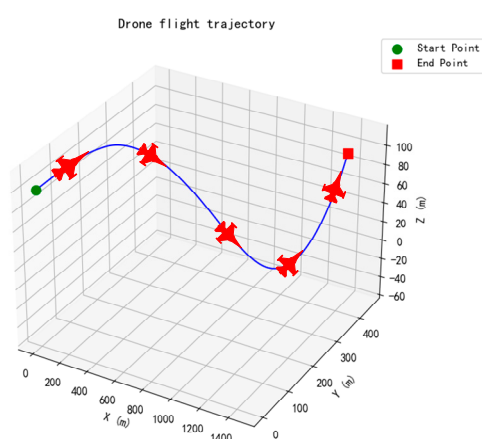


Figure 6. Drone flight trajectory.

For k maneuver categories, a total of k binary-classification SVM models are trained. Each model distinguishes this category from all other categories. Specifically, for the i -th category ($i = 1, 2, \dots, k$), when training the corresponding binary classifier, the samples of this category are marked as positive examples, and the samples of all other categories are uniformly marked as negative examples. In this way, each classifier focuses on learning the decision boundary between the target category and other categories.

In addition, to adapt to the probabilistic analysis of UAV maneuver intentions, the Platt Scaling method is used to perform nonlinear fitting on the decision function output of SVM. A sigmoid function is introduced, and its parameters A and B are adjusted through maximum likelihood estimation to convert the discrete classification results into continuous probability values. The formula is:

$$P(y = 1|x) = \frac{1}{1 + \exp(Af(x) + B)} \quad (15)$$

Here, $f(x)$ is the decision function of SVM, which is generally solved by maximum likelihood estimation. After being mapped by the Sigmoid function, the output value falls in the interval $(0, 1)$, which can be interpreted as “the probability that the sample belongs to the positive example category (such as ‘climb’)”. When the result outputs of all classifiers are obtained, the category with the maximum selection probability is taken as the probability distribution of each type of maneuver action, providing more quantitatively valuable decision-making basis for UAV intention recognition.

The workflow of the UAV flight intention recognition model is shown in Figure 7. The random forest algorithm is used to perform feature screening on high-dimensional flight state information. The original features of flight information are evaluated through information gain, and the core features that contribute significantly to classification are selected. A multi-class SVM model with the OvA strategy is adopted. A dedicated binary classifier is trained for each category to learn the boundary between this category and all other categories, providing information for the subsequent trajectory prediction model.

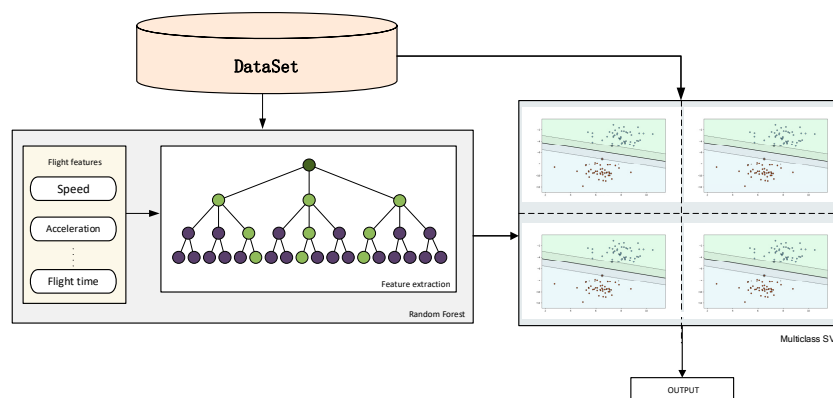


Figure 7. SVM algorithm process.

4. Design of UAV Trajectory Prediction Algorithm

This section mainly introduces the trajectory prediction algorithm based on neural networks. Section 4.1 introduces the Bi-GRU neural network. With the help of a bidirectional processing mechanism, this network can capture the dependency information of the sequence in both forward and backward directions simultaneously. Section 4.2 introduces the principle of the attention mechanism, and adopts the Bi-GRU encoding-decoding structure, fusing the maneuver intention probability distribution output by SVM, to construct a complete process from flight state perception to accurate trajectory prediction, and realize the effective prediction of UAV trajectories.

4.1. Bi-GRU Algorithm

Bi-GRU, as an important variant of RNN [34], is different from LSTM in architecture design. Its core advantage lies in simplifying the network structure through two gating units, namely the Reset Gate and the Update Gate. While effectively alleviating the gradient vanishing problem, it significantly improves the calculation efficiency, making it more suitable for scenarios with real-time requirements such as UAV trajectory prediction.

For a single-layer GRU network, the Reset Gate is used to control the degree of association between the input information at the current time step and the hidden state at the previous time step. It determines how much information is transferred from the hidden state at the previous time step to the calculation at the current time step. Its formula is:

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \quad (16)$$

Here, r_t is the output of the reset gate at time step t , σ is the activation function of the network, x_t is the input at the current time step, h_{t-1} is the hidden state at the previous time step, and W_r , U_r , b_r are the weights and biases of the reset gate.

When r_t is close to 0, it indicates that the input information at the current time step almost completely comes from the updated information of the hidden state at the previous time step, and the hidden state at the previous time step is “reset” or “ignored”.

The Update Gate mainly controls the “memory” degree when updating the hidden state at each time step, that is, how much of the hidden state at the current time step comes from the hidden state at the previous time step and how much comes from the input information at the current time step. Its formula is:

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \quad (17)$$

Here, z_t is the output of the update gate at time step t , W_z , U_z , b_z are the weights and biases of the update gate.

In Bi-GRU, the hidden state h_t at each time step is the result of the joint action of the Reset Gate and the Update Gate. Specifically, the update process can be expressed as follows:

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (18)$$

Here, $\odot$ represents element-wise multiplication.

Bi-GRU consists of two independent GRU layers, namely the forward layer and the backward layer. When processing sequence information, the forward layer processes according to the input order, while the backward layer processes in the reverse order, so as to capture the bidirectional time dependence of the sequence. As shown in Figure 8, for an input sequence $X = [x_1, x_1 \cdots x_T]$, the propagation process of the Bi-GRU model can be described as:

$$\vec{h}_t = \text{GRU}(\vec{h}_{t-1}, x_t) \quad (19)$$

$$\overleftarrow{h}_t = \text{GRU}(\overleftarrow{h}_{t+1}, x_t) \quad (20)$$

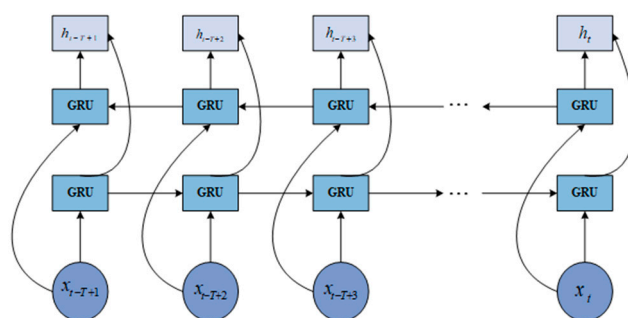


Figure 8. Bi-GRU algorithm.

In the formulas, $\vec{h}$ is the hidden state from left to right, $\overleftarrow{h}$ is the hidden state from right to left; x_t is the t -th element of the input sequence. The final hidden state is obtained by concatenating the hidden states in the two directions, as shown in the following formula:

$$h_z = \left[\vec{h}_t; \overleftarrow{h}_t \right] \quad (21)$$

Input the fused hidden state into a fully connected layer, and output the prediction result y_t after being activated by *Softmax*. The formula is:

$$y_t = \text{Softmax}(Wh_t + b) \quad (22)$$

In the formula, W is the weight of the fully connected layer, and b is the bias of the fully connected layer.

To quantify the prediction error, the loss function adopts the Mean Squared Error (MSE), which measures the difference between the predicted position $\hat{y}_t$ and the real position y_t . Its formula is:

$$\Gamma = \frac{1}{N} \sum_{t=1}^N \|y_t - \hat{y}_t\|^2 \quad (23)$$

N is the number of training samples.

Compared with the unidirectional GRU, Bi-GRU, with the ability to capture bidirectional time dependence, can more comprehensively excavate the temporal correlation in sequence data, which is in line with the demand for dynamic information perception in the real-time trajectory prediction of UAV.

4.2. Attention-Bi-GRU Algorithm

The attention mechanism is an adaptive information processing framework [35]. It dynamically focuses on the key content of sequence problems, calculates the importance weights of parts, and extracts the key information of time-series information. This paper constructs a Bi-GRU encoding-decoding architecture, fuses the attention mechanism, and realizes the real-time prediction of UAV trajectories. The overall network framework is shown in Figure 9.

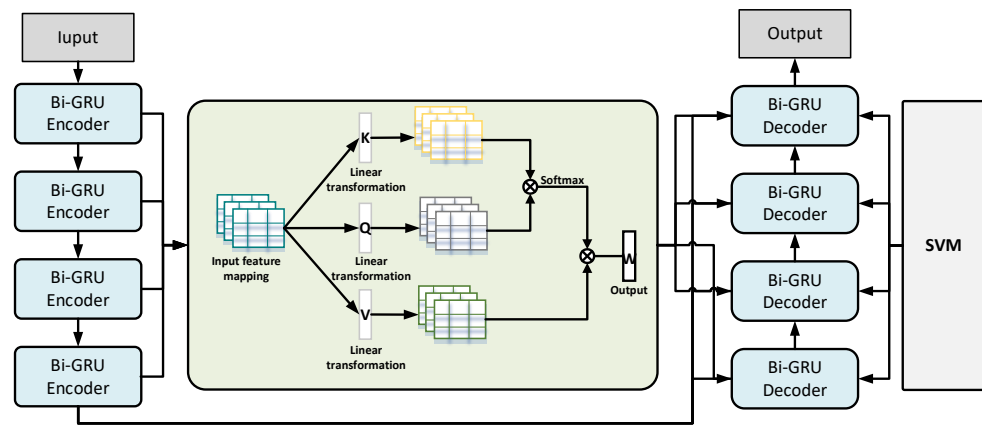


Figure 9. The Bi-GRU algorithm based on Attention.

4.2.1. Encoder

The encoder is responsible for converting the input sequence into a high-dimensional context representation. This paper adopts a multi-layer encoder structure. Each layer includes a Feed-Forward Neural Network (FFNN) and a Bi-GRU neural network layer, as shown in Figure 10. Assume that the given input sequence $X = [x_1, x_2, \dots, x_n]$, where $x_t \in R^d$ is the feature vector of the trajectory at time t . First, the original input is subjected to feature enhancement and nonlinear transformation through the feed-forward neural network:

$$\tilde{x}_t = FFN(x_t) = \text{ReLU}(W_1 x_t + b_1) W_2 + b_2 \quad (24)$$

Here, $W_1 \in R^{d \times h_1}$, $W_2 \in R^{h_1 \times h_2}$ are weight matrices; $b_1 \in R^{h_1}$, $b_2 \in R^{h_2}$ are bias vectors. h_1 , h_2 are the dimensions of the hidden layer and the output layer, respectively; ReLU is the activation function.

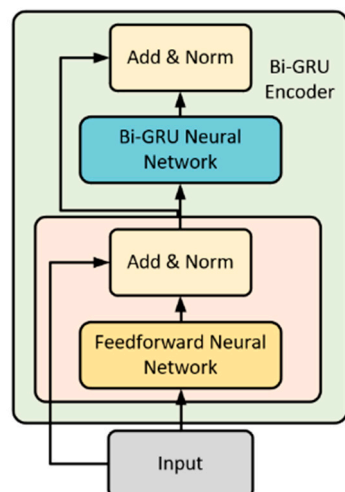


Figure 10. Bi-GRU encoder.

The feature information enhanced by FFN is input into the Bi-GRU network, and the context information of the input sequence is extracted from the time-series features, and the hidden state of each time step is updated. The specific calculation formula is as follows:

$$h_t = Bi - GRU(h_{t-1}, \tilde{x}_t, W_1, W_2) \quad (25)$$

In the formulas, h_t represents the output of the hidden state. The bidirectional structure of Bi-GRU ensures that information can be transmitted from the past and the future simultaneously.

To improve the stability and training efficiency of the model, an Add & Norm structure is designed between each layer of the neural network. With the help of Residual Connection, the network input and output are directly added to construct a skip connection, realizing cross-layer transmission and avoiding gradient disappearance. It is combined with Layer Normalization to standardize the feature distribution and accelerate model convergence. The formula of Layer Normalization is as follows:

$$LayerNorm(x) = \frac{\tilde{x} - \mu}{\sigma} \times \gamma + \beta \quad (26)$$

Here, γ and β are the scaling factor and the bias term.

4.2.2. Attention Mechanism Layer

The attention mechanism layer is the core computing unit of this section. Using the hidden states from the encoding layer as Key (K) and Value (V), and the current decoding requirement as Query (Q), we compute adaptive weights through Q-K-V interaction. This distills the entire historical trajectory into a context vector that best aligns with the maneuver intention, directly mapping encoded semantics to trajectory prediction.

In the trajectory prediction model based on Bi-GRU, Q, K, and V are all generated from the hidden state through linear transformation, but there are essential differences in the input source and function.

h_t^d is the hidden state of the current moment output by the Bi-GRU decoder. The prediction intention is mapped to the attention interaction space through the W_Q weight matrix. Its formula can be expressed as:

$$Q = \eta(h_t^d, W_Q) \quad (27)$$

h_i^e is the hidden state at the i -th moment output by the Bi-GRU decoder. The historical trajectory features are, respectively, mapped as index and content values through the W_K and W_V weight matrices. Its formula can be expressed as:

$$K = \rho(h_i^e, W_K) \quad (28)$$

$$V = \delta(h_i^e, W_V) \quad (29)$$

The core of the Attention mechanism is to calculate the similarity between the Q value and the V value, dynamically weight the V value, and obtain the attention weight vector α through the softmax function normalization to determine which part of the input sequence the decoder should focus on. The formula is as follows:

$$\alpha = \text{softmax}\left(\frac{QK^T}{\sqrt{d_K}}\right)V \quad (30)$$

Here, α is the output weight, T is the sequence length, d_K is the feature dimension. $\frac{QK^T}{\sqrt{d_K}}$ represents the dot product of the Q value and the K value, reflecting the matching degree between the current prediction and the historical trajectory.

4.2.3. Decoder

The decoder adopts a multi-layer cascaded structure. Each layer of the decoder is consistent with the encoder in architecture design. Each layer of the decoder receives two key inputs: one is the focused feature α output by the attention mechanism layer, and the other is the maneuver probability distribution P_S output by the SVM maneuver intention recognition model. First, for the probability distribution output by SVM, which is low-dimensional semantics, a linear transformation is used to align it with the Bi-GRU encoding. The formula is expressed as:

$$z_S = \text{GeLU}(W_P P_S + b_P) \quad (31)$$

Here, GeLU is the activation function, W_P and b_P are the weight and bias term of the linear transformation.

The concatenation operation is adopted to fuse the focused feature output by the attention mechanism layer and the transformed SVM feature z_S . The formula is expressed as:

$$h = \text{Concat}(\alpha, z_S) \quad (32)$$

5. Experiments and Results

This section is divided into two parts. Section 5.1 introduces the hardware and software platform of the experiment and the dataset, and Section 5.2 designs three types of experiments to verify the effectiveness of the proposed algorithm framework.

5.1. Construction of Experiment Platform and Dataset

The hardware and software platform used in the experiment is listed in Table 1.

To comprehensively cover both basic and composite maneuvers encountered in real UAV operations, this study employs a flight-track dataset generated in DCS World. Eight flight missions were executed, yielding a total of 100 continuous trajectories. According to experimental requirements, these trajectories were segmented into 320 sample units: 120 basic-maneuver samples and 200 composite-maneuver samples. The dataset encompasses typical flight patterns such as straight flight, acceleration, level turns, and climbing turns. The construction procedure is as follows:

Table 1. Experimental software/hardware configuration.

Hardware	Configuration	Software	Configuration
CPU	Intel i9-14900KF	Programming language	Python 3.8
GPU	NVIDIA GeForce RTX 4090D	Deep learning framework	Pytorch 2.2
Memory	DDR5 5600 MHz 16 GB x4)	Third-party libraries	Numpy 1.18.x; Scikit-learn etc.
Operating system	Windows 10	IDE	Pycharm

5.1.1. Basic-Maneuver Samples

For each of the four basic maneuvers—level flight, turning, climbing, and diving—30 independent samples were collected, each lasting ≥ 10 s. A multi-label scheme is adopted: if a sample contains a specific maneuver, the corresponding label is set to 1; otherwise, it is set to 0.

5.1.2. Composite-Maneuver Samples

From the entire trajectory pool, a sliding window with a fixed length of 5 s and a step size of 1 s is applied, generating 200 trajectory snippets. Each snippet is annotated with a multi-class label according to the true maneuver combination it contains.

5.1.3. Data Isolation and Partitioning

Basic- and composite-maneuver samples are completely isolated in terms of trajectory IDs and time intervals. All samples are randomly split into training and testing sets at an 8:2 ratio after global shuffling, ensuring no data leakage.

5.2. Simulation Experiment Verification

5.2.1. Maneuver Intention Experiment Verification

In the training stage, the SVM selects the Radial Basis Function (RBF) as the baseline classifier, and uses grid search to optimize the penalty coefficient C and the kernel width γ . $C = 5$ and $\gamma = 0.01$ are determined. To eliminate the dimension difference in data, all features are normalized to the $[0, 1]$ interval. The evaluation indicators select Precision, Recall, and F1-Score, which are defined as follows:

$$P = \frac{TP}{TP + FP} \quad (33)$$

$$R = \frac{TP}{TP + FN} \quad (34)$$

$$F_1 = 2 \times \frac{P \times R}{P + R} \quad (35)$$

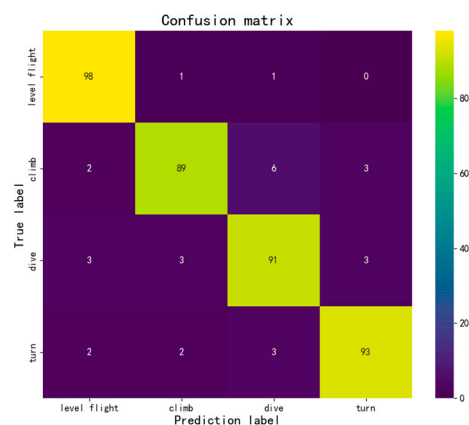
Among them, TP is the true positive count, FP is the false positive count, and FN is the false negative count.

Table 2 presents the classification results of the four types of maneuvers. It can be seen that the overall performance of the model is excellent, with an average precision of 0.96, a recall rate of 0.96, and an F1-score of 0.95, indicating that a good balance has been achieved between “precise recognition” and “comprehensive coverage”. Among them, the F1-scores of level flight and dive reach 0.98 and 0.95, respectively, verifying the robustness of the algorithm for a single maneuver; the indicators for turn and climb are slightly lower but still remain above 0.92, indicating that it also has stable recognition ability for composite maneuvers.

Table 2. The classification results of the four types of maneuvers.

Category	Precision	Recall	F1-Score
Level flight	0.99	0.98	0.98
Turn	0.95	0.89	0.92
Climb	0.93	0.91	0.92
Dive	0.98	0.93	0.95
Average	0.96	0.93	0.95

Use a confusion matrix (as shown in Figure 11) to drill down into the misclassification details. The high proportion of diagonal elements shows that most samples are correctly classified; the off-diagonal errors are mainly concentrated in the cross-region of turn and dive. The reason may be that there is partial overlap in the vertical acceleration features of the two types of maneuvers, leading to a very small number of misjudgments. Overall, the average recognition accuracy of SVM exceeds 0.96, fully proving the effectiveness and reliability of the method in this paper for the task of UAV maneuver intention recognition.

**Figure 11.** Confusion matrix.

The classification results show that the SVM model has a relatively high accuracy in the recognition of UAV maneuver intentions, verifying that the algorithm proposed in this section has a certain effect.

5.2.2. Trajectory Prediction Experiment Verification

In this experiment, multi-dimensional trajectory features—comprising 3-D position, velocity, acceleration, attitude angles, their first derivatives, and height rate—are fed into the network to predict the UAV's 3-D Cartesian coordinates over the next five time steps. Hyper-parameters are optimized via Optuna's Bayesian search, yielding the configuration listed in Table 3. To make full use of hardware resources and improve computing efficiency, the training and testing processes of the model are both executed in the aforementioned GPU environment.

Table 3. Experimental parameter settings.

Parameter	Value	Parameter	Value
Number of time windows	16	Dropout	0.2
Number of Bi-GRU network layers	2	Learning rate	0.0005
Number of hidden units per Bi-GRU	64	Batch Size	64
Number of feed—forward network layers	2	Optimizer	Adam
Number of decoders/encoders	4	Epochs	300

Figure 12 illustrates the evolution of the loss on both the training and the test sets. Overall, the loss decreases rapidly at the beginning and then gradually levels off, demonstrating stable convergence. Specifically, during the first 50 epochs, the steep descent indicates that the model quickly captures the dominant patterns. Between epoch 50 and 150, the slower reduction suggests fine-grained feature learning. After epoch 150, the curve flattens and stabilizes around 0.002, confirming that the network has achieved an optimal convergence and has been sufficiently trained.

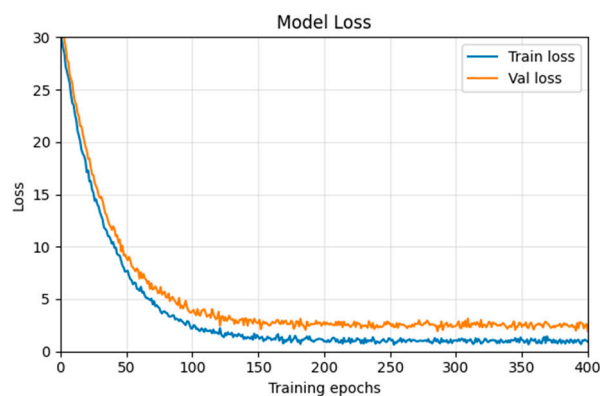


Figure 12. Loss function change curve.

To quantitatively evaluate the performance gain of the proposed Attention-Bi-GRU algorithm in trajectory prediction, this section conducts a comparative experiment against mainstream temporal neural networks. Three models are examined under an identical dataset and unified metrics: (i) a classical encoder–decoder architecture, (ii) a standard LSTM network, and (iii) the proposed Attention-Bi-GRU fusion model. The analysis focuses on the differences in short-term dynamic capture and long-term trajectory evolution.

Three-dimensional visualizations of the predicted trajectories are provided to inspect the 3-D morphology, axis-wise dynamics (x , y , z), and local trend characteristics. Figure 13 presents the comparative results: the blue curve denotes the actual UAV flight path, the red curve the encoder–decoder prediction, the green curve the LSTM prediction, and the yellow curve the proposed Attention-Bi-GRU prediction.

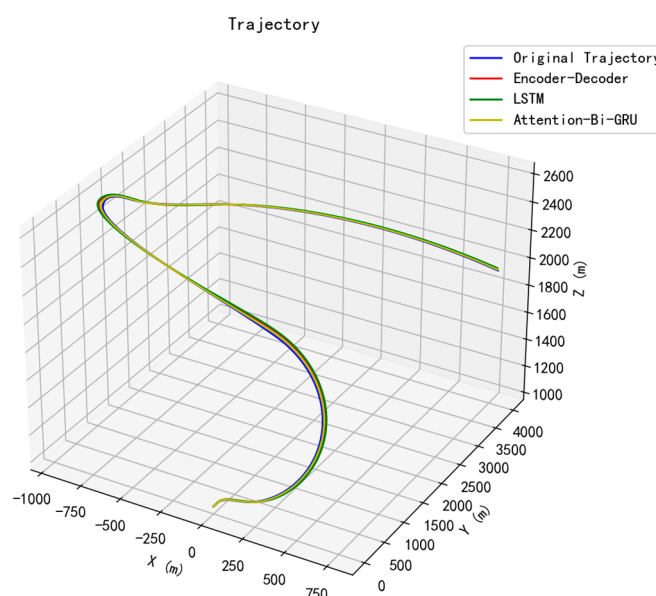


Figure 13. Comparison chart of 3D track simulation prediction.

The flight segment from 30 to 35 s is magnified in 3-D and along the X-axis (see Figure 14). During this window the true trajectory exhibits pronounced nonlinear maneuvers, imposing stringent requirements on short-term dynamic modeling. The Attention-Bi-GRU model (yellow) closely tracks the real path, demonstrating high fidelity in capturing abrupt changes. By leveraging bidirectional temporal context and attention-weighted key time-steps, the model effectively suppresses error accumulation. In contrast, both the Bi-GRU (red) and the encoder–decoder (green) under-represent the local X-axis variations, failing to reproduce the fine fluctuations and consequently incurring larger local errors, which reveals their limited expressiveness under high-dynamic conditions.

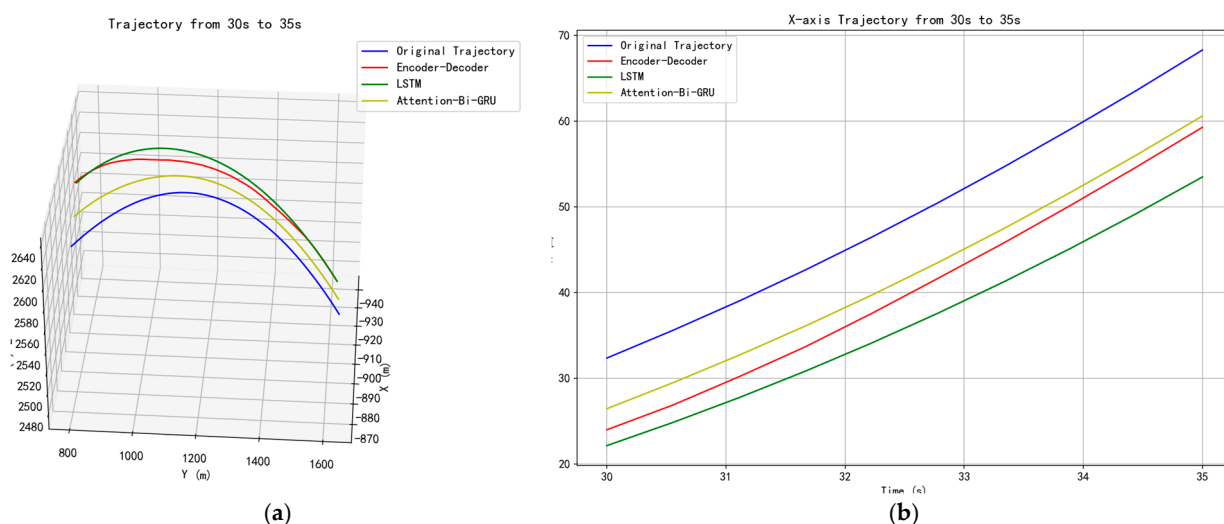


Figure 14. 30 s–35 s local magnification. (a) 3D local magnification. (b) X-axis local magnification.

Relative to the aggressive turn in Figure 14, the 25–30 s segment involves only minor heading corrections. As illustrated in Figure 15, Attention-Bi-GRU (yellow) still adheres tightly to the true trajectory. This consistent accuracy in low-dynamic scenarios confirms the model's robustness and generalization capability for small-amplitude maneuvers.

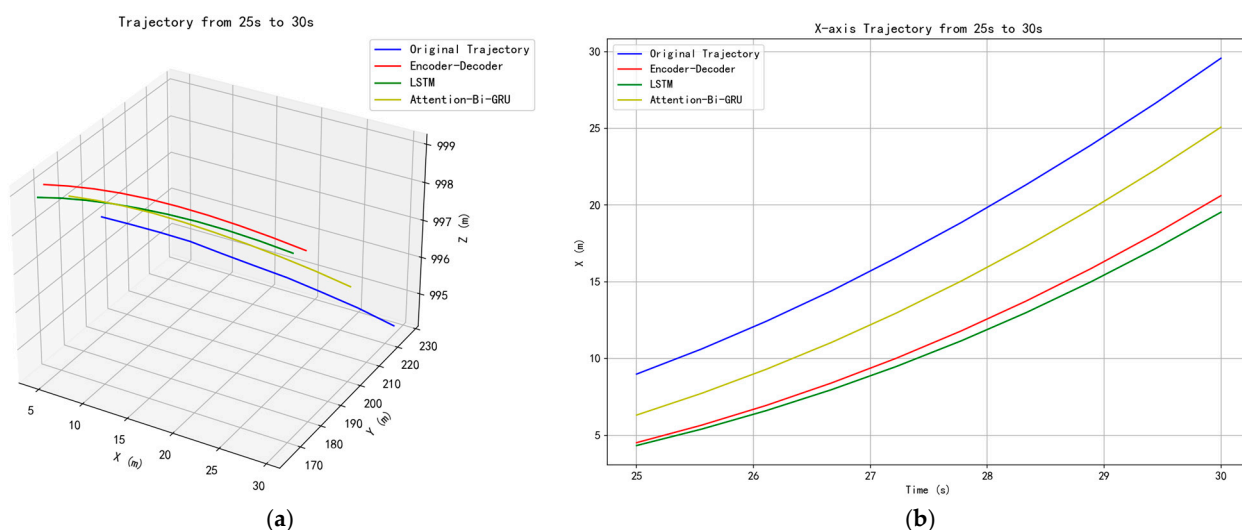


Figure 15. 25 s–30 s local magnification. (a) 3D local magnification. (b) X-axis local magnification.

In the qualitative analysis, The Cumulative Average Euclidean Distance Error (CAEDE) for each model as a function of time t is computed, as shown in Figure 16. The Attention-Bi-GRU model (yellow line) exhibits the smallest slope throughout the entire interval; at

50 s its CAEDE is approximately 10 m, significantly lower than that of the other models. The LSTM (green line) presents the steepest slope during 0–20 s and eventually reaches 25 m, indicating insufficient long-term dependency modeling and a “solidified” early error that keeps amplifying. The encoder–decoder (red line) grows sharply in 0–10 s; although its slope decreases slightly after 10 s, it remains higher than the steady-state level of the yellow line, so the model cannot effectively “stop the loss” later on. These CAEDE results robustly verify the sustained accuracy and resilience of the proposed Attention-Bi-GRU algorithm in trajectory prediction tasks.

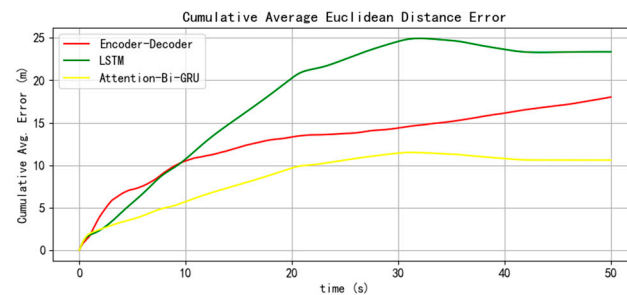


Figure 16. The cumulative average euclidean distance error.

The three-dimensional Root Mean Square Error (RMSE) and Trajectory Overlap are calculated for all three models to quantify prediction accuracy and shape consistency, as listed in Table 4. RMSE reflects the average 3-D deviation between predicted and true trajectories; smaller values denote higher positioning accuracy. Overlap is defined as the spatial Intersection over Union (IoU) between predicted and ground-truth trajectories within the same time window, with a coincidence threshold set to 15 m; its range is [0, 1], where values closer to 1 indicate higher coincidence.

Table 4. Indicator comparison.

Parameter	RMSE(X)	RMSE(Y)	RMSE(Z)	Overlap(X)	Overlap(Y)	Overlap(Z)
LSTM	15.73	15.89	11.42	56.6%	62.8%	83.0%
Encoder/Decoder	12.71	12.88	9.15	89.2%	82.8%	92.6%
Attention-Bi-GRU	6.97	7.22	5.17	100.0%	100.0%	100.0%

Table 4 shows that the proposed Attention-Bi-GRU achieves significantly lower RMSE than the Encoder–Decoder and LSTM networks in the X, Y and Z dimensions, and its Overlap reaches 100%. These results demonstrate that the proposed algorithm not only maintains high positional accuracy but also better reproduces the overall shape and detailed maneuvers of the true trajectory.

5.2.3. Trajectory Prediction Algorithm Based on Intention Recognition

This section introduces the intent recognition module in the trajectory prediction network and conducts comparative experiments under the exact same conditions. The results are shown in Figure 17. The blue curve is the actual trajectory of the UAV; the green curve is the predicted trajectory of the original network without fused intention information; the red curve is the predicted trajectory of the improved network after fusing intention information. By comparison, it can be seen that the introduction of intention information significantly improves the prediction precision, making the green trajectory closer to the actual trajectory, verifying the effectiveness of the intention enhancement strategy.

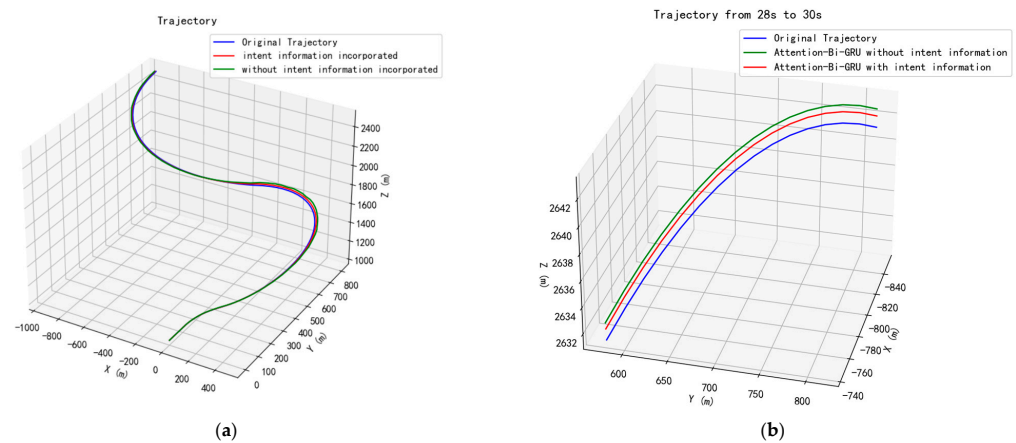


Figure 17. Comparison of before and after fusion intent information. (a) 3D local magnification. (b) X-axis local magnification.

From the simulation results, the model shows high accuracy in trajectory prediction in both scenarios. In Figure 17a, the actual trajectory and the predicted trajectory after fusing intention information almost completely overlap, indicating that the model can accurately capture the movement trend and spatial characteristics of the target. The predicted trajectory without fused intention information has a consistent trend with the actual trajectory, but there are still slight deviations. This indicates that the introduction of intention information significantly enhances the model's representation ability for dynamic trajectories, enabling the model to more accurately predict the future position of the target. In Figure 17b, the 28 s to 30 s local area is restricted. It can be seen that in the peak area, the model's dynamic capture ability for the target trajectory is fully demonstrated. In contrast, the prediction error of the trajectory without fused intention information is slightly higher.

To further verify that the effect of the Attention-Bi-GRU model fusing target intention information is better than that without fused intention information, this section will conduct quantitative analysis. The Root Mean Square Error (RMSE) is used as a performance evaluation index to accurately evaluate the trajectory simulation results. The error comparison before and after fusing intention information is shown in Table 5. By calculating these indexes, the error level of the model in the trajectory prediction task can be more objectively measured.

Table 5. RMSE comparison of two models.

Parameter	X	Y	Z
Without fused intention module	8.25	8.47	5.93
With fused intention module	4.07	4.36	2.93

The simulation results show that after fusing intention information, the errors of the model in all dimensions are significantly reduced. It indicates that fusing intention information can make the model better capture the movement laws of UAVs, effectively making up for the model's deficiencies in the global movement trend and complex trajectory characteristics. In terms of performance in various dimensions, the error in the X dimension is the lowest, indicating that its movement law is relatively simple, and the model can achieve good prediction results even without fused intention information; while the errors in the Y and Z dimensions are relatively high, and the decline after fusing intention information is significant, indicating that intention information plays a key role in capturing complex trajectory changes. It provides powerful support for further optimizing the UAV trajectory prediction model.

6. Conclusions

In this paper, the UAV maneuver intention is embedded into the trajectory prediction process, and a set of “intention recognition—trajectory prediction” collaborative framework is constructed. A random forest-SVM hybrid model is used to extract key features from redundant high-dimensional flight states, and the probabilistic maneuver intention provides constraints for the trajectory network. An Attention-Bi-GRU network is designed, and the bidirectional spatio-temporal representation and attention focusing mechanism are used to accurately capture the trajectory evolution law in complex environments. Comparative experiments indicate that injecting intention information decreases the average RMSE by over 45% and significantly enhances prediction precision for flight states. This framework is expected to further evolve into a prediction engine with autonomous decision-making, conflict resolution, and traffic flow optimization in dynamic airspace in the future as the deep learning theory of the next generation of flight control systems continues to advance, providing reliable, transplantable, and sustainable upgrade capabilities. In future work, we will concentrate on multi-UAV cooperative scenarios, leveraging additional state-of-the-art network architectures to further unlock the performance limits of the intention–trajectory coupling framework, thereby achieving more accurate and robust trajectory prediction.

Author Contributions: Conceptualization, Y.C. and J.Z.; methodology, Y.C.; software, Y.C. and C.Z.; validation, Y.C., J.Z. and Q.Y.; formal analysis, Y.C.; investigation, Y.C.; resources, G.S.; data curation, Y.C.; writing—original draft preparation, Y.C.; writing—review and editing, J.Z. and Q.Y.; visualization, Y.C.; supervision, G.S. and Q.Y.; project administration, G.S.; funding acquisition, G.S. and Q.Y. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the Aeronautical Science Foundation of China (20220013053005).

Data Availability Statement: The original contributions presented in the study are included in the article, further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest. Author Chengbiao Zhang was employed by the company Jiangxi Hongdu Aviation Industry co., Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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